GRIDPRINT analysis report

Verdict

✓

VERIFIED

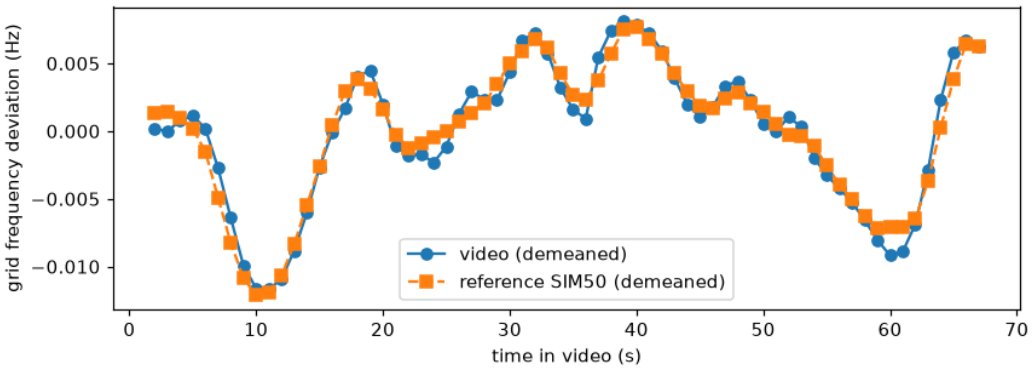
A grid signal is present and matches the reference log at one time. No significant edit evidence was found.

Report state: FINAL

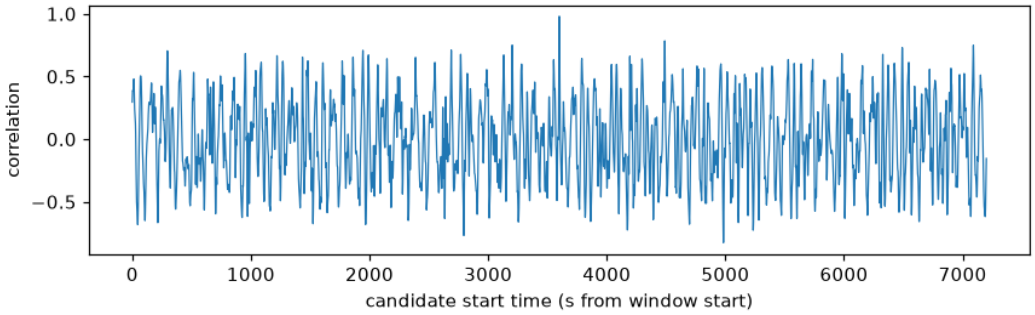
Summary

Estimated recording start (UTC)	2026-01-20T10:00:00.014252+00:00 ± 1.5 s
Grid zone	SIM50
Reasons	• `READOUT_FROM_SIDECAR` — Readout time came from the Capture sidecar. • `MATCH_OK` — The track matches the reference log at one time. • `DROPPED_FRAMES_EXPLAINED` — Frame gaps were found; the flicker phase across them is consistent (not an edit).
Match	peak correlation 0.980, next peak 0.784, track length 69 s

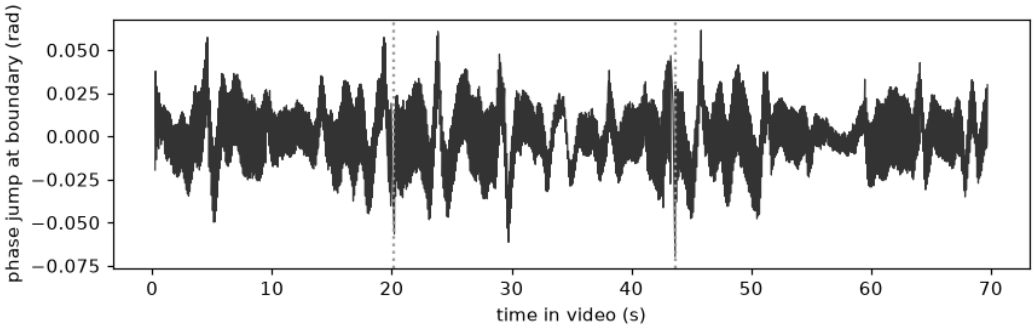
Key plots



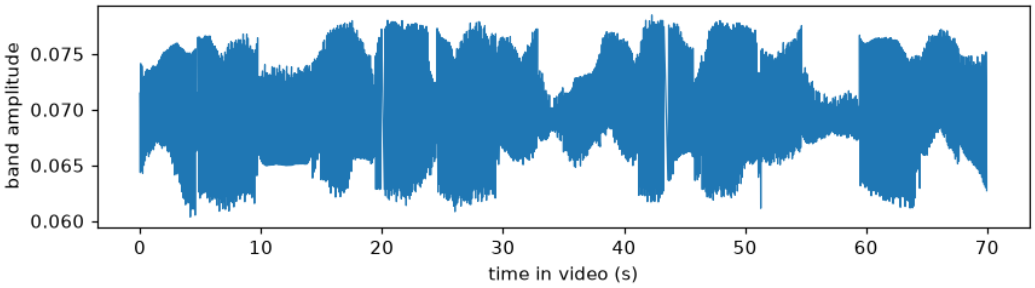
Measured grid frequency variation and matched reference.



Correlation against candidate start times.



Phase jump at frame boundaries. Solid line: edit candidate. Dotted line: timestamp gap.



Flicker band amplitude per frame (fractional modulation).

Limitations

- GRIDPRINT reads the grid frequency from the flicker of mains-powered lights. It is not a deepfake classifier and it is not cryptographic proof.
- A missing signal is never evidence that a video is fake. Many phones hide the flicker ("anti-banding"), and daylight or battery lights do not flicker.
- A time estimate is possible only inside the time range of the reference log for the zone.
- Location output is limited to the grid zone. GRIDPRINT gives no finer location.
- Thresholds come from calibration data. Results outside the calibrated conditions carry more risk of error.
- A CONTRADICTED verdict needs approval by a human reviewer before the report is final.

Provenance

File SHA-256	`97bd81ea29c2a33b16465d912f7ac4e526a0425b1cc54b33e27aeb10b46074b9`
Engine version	0.1.0
Config hash	`8f439372b865ee8e`
Thresholds hash	`5c105287d4e1c479`
Job	`aws3-dropped-frames-040218`

Appendix

Continuity events

Kind	Time in video (s)	Phase jump (rad)	Significance
DROPPED_FRAMES_EXPLAINED	20.20	-0.033	0.6

DROPPED_FRAMES_EXPLAINED	43.63	0.020	0.3

Quality metrics

axis	rows+
band_snr_db	11.426580954399864
bitrate	490747
bits_per_pixel	0.21453018834932053
camera_motion_px	0.032422965009069944
codec	h264
dft_peak_hz	100.02934814260769
estimators_agree	True
foreground_fraction	0.052121927458033575
frames	2085
height	240
jitter_p50_ms	3.3305269653283176e-07
jitter_p95_ms	3.338056941970535e-07
motion_level_px	0.10989784047152439
nominal_flicker_hz	100.0
opencv_frame_count	2085
phase_mean_flicker_hz	100.0261382279723
readout_s	0.025
readout_source	READOUT_FROM_SIDECAR
rotation_metadata_deg	0
segmentation	ok
static_fraction	1.0
timestamp_source	sensor
timing_assumption	sensor_timestamp_is_first_row_exposure_start
track_id	t1
track_samples	66

transfer	bt709
verdict_rule	S13.4
width	320

Stage timings (s)

S1_ingest	0.2987
S2_linear	2.1473
S3_readout	0.0
S4_motion	6.429
S5_segment	0.9464
S7_rowsignal	1.3119
S6_patches	1.5387
S8_phase	0.7588
S10_match	0.6712
S11_continuity	0.0009